YEAR 7 · SCIENCE 3.4–3.6 · MATHS 5.1 · EXTRA HOMEWORK 11

How High Does It Bounce?

Tides · bouncing balls · joules · a cooling coffee · angles — all in decimals

Name: _____ **Class:** _____

Date given: _____ **Date due:** Friday 6 November 2026

Why this sheet exists

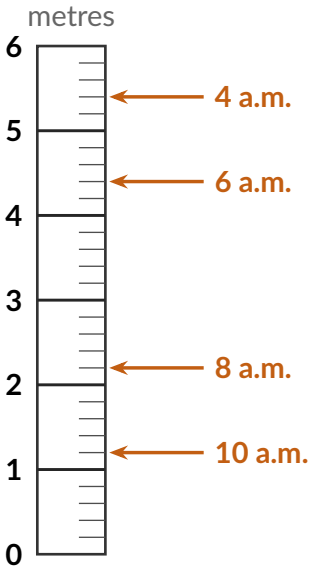
There is **nothing new to calculate** here. Every number is a Unit 4 decimal, and every question is this week's science: the tides, energy, and a ball that will not bounce back to where it started.

The steps are printed as **boxes**. Fill the boxes — that *is* the method. An **orange box** means: write a **0** here.

No calculator. Answer in **full sentences** when it says so.

Part 1 · Tides in decimals

X1 Read the tide board



Word help

tide board	a tall ruler on the harbour wall. It shows the **depth** of the sea.
depth	how deep the water is

Mr Bowen looks at the tide board every 2 hours. Every small line is **0.2 m**. Count in 0.2s: 5, 5.2, 5.4.

(a) Write the depth at each time.

Time	4 a.m.	6 a.m.	8 a.m.	10 a.m.
Depth (m)				

(b) Which time was **high tide**? _____

(c) Which time was **low tide**? _____

X2 The tidal range

Tidal range, from Science 3.4

Tidal range: the difference in depth between high tide and low tide. It is a **subtraction**: high tide − low tide.

Write a 0 in every orange box. Then subtract. **(a)** is done for you.

(a) Mr Bowen's harbour
high 5.4 m, low 1.2 m

	5	.	4	0
−	1	.	2	0
	4	.	2	0

Range: **4.2 m**

(b) Another harbour
high 3.75 m, low 0.9 m

	3	.	7	5
−	0	.	9	0
		.		

Range: _____ m

(c) A third harbour
high 6.1 m, low 2.35 m

	6	.	1	0
−	2	.	3	5
		.		

Range: _____ m

(d) The Bay of Fundy in Canada has the biggest tides in the world. Its tidal range is **16.3 m**. How much bigger is that than the range at Mr Bowen's harbour (4.2 m)?

how much bigger? = the difference = subtract

Line up the points in the grid yourself.

Answer: _____ m

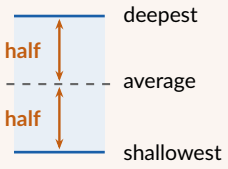
		.	
−		.	
		.	

X3 Half above, half below

The deepest water

The water goes **half the range above** the average depth, and **half the range below** it.

Average 5.0 m, range 2.0 m: half the range is 1.0 m.
Deepest: 5.0 + 1.0 = **6.0** m. Not 7.0 m!



Fill in the table. The first row is done for you.

Average depth	Tidal range	Half the range range ÷ 2	Deepest average + half	Shallowest average − half
5.0 m	2.0 m	2.0 ÷ 2 = 1.0	5.0 + 1.0 = 6.0	5.0 − 1.0 = 4.0
4.5 m	3.0 m	3.0 ÷ 2 =	4.5 +	4.5 −
3.2 m	1.4 m			
6.0 m	2.5 m			

(b) Mr Bowen says: *"In row 2, the deepest water is* 4.5 + 3.0 = 7.5 *m."* What did he do wrong? Answer in a full sentence.

X4 The land has tides too

The land goes up and down too: an **earth tide**. Hanoi goes up and down about **30 cm**, twice a day.

(a) 100 cm = 1 m, so divide by 100. 30 cm = _____ m

(b) Hanoi goes **up** that far twice a day. How far up is that in one day?

0.3 × 2 = _____ m

(c) How far up does Hanoi go in **one week** (7 days)? (It comes down again every time.)

Point out: 6 × 7 = _____ Point back: 0.6 × 7 = _____ m

(d) Nobody in Hanoi feels it. Why not? Answer in a full sentence. Start: **"We do not feel it because …"**

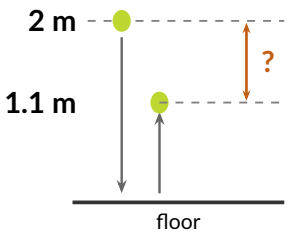
Part 2 · Energy in decimals

X5 Six balls, one floor

Mr Bowen drops six balls, one at a time. He drops each one from **2 m**. He measures how high it **bounces**.

(a) Fill in the last column. The first row is done for you.

Ball	Bounce	How far below 2 m?
tennis ball	1.1 m	2.00 − 1.10 = 0.90 m
ball of clay	0.05 m	2.00 − 0.05 =
bouncy ball	1.75 m	2.00 − 1.75 =
an egg	0 m	2.00 − 0 =
basketball	1.35 m	2.00 − 1.35 =
ping-pong ball	1.5 m	2.00 − 1.5 =



Word help

How far below 2 m? = the difference = **subtract**.
Write 2 as **2.00**. Or count up to 2: 1.6 + 0.4 = 2.

(b) The six bounces were 1.1, 0.05, 1.75, 0, 1.35 and 1.5 m. Write them in order. Start with the **highest**.

_____ _____ _____ _____ _____ _____

(c) At the bounce, some energy changes to **sound** and **heat**.

Which ball changed the **most** energy to sound and heat? _____

Which ball changed the **least**? _____

(d) Finish the sentence.

The lower the bounce, the more energy …

X6 Drop it three times

The mean

Step 1: add up all the values. **Step 2:** divide by how many there are. Three drops: divide by **3**.

(a) The tennis ball bounces **1.12 m**, **1.08 m** and **1.13 m**. Find the mean.

Step 1 add

$$\begin{array}{r} \square 1.12 \\ \square 1.08 \\ +\ \square 1.13 \\ \hline \square\square.\square\square \end{array}$$

Step 2 ÷ 3

$$3 \overline{)\ 3.33}$$

Mean: _____ m

(b) The basketball bounces **1.34 m**, **1.38 m** and **1.36 m**. This time, write the digits in the grid yourself. Then write the total in the division.

Step 1 add

$$\begin{array}{r} \square\square.\square\square \\ \square\square.\square\square \\ +\ \square\square.\square\square \\ \hline \square\square.\square\square \end{array}$$

Step 2 ÷ 3

$$3 \overline{)\ \square\square.\square\square}$$

Mean: _____ m

X7 Joules

The book's numbers, from Science 3.5

1 floor of stairs: **2 000 J** run 1 metre: **200 J** boil 1 litre of water: **400 000 J**

(a) The kettle. 1 litre needs 400 000 J, so 0.1 litre needs 400 000 ÷ 10 = 40 000 J. Fill in the table.

Water (litres)	How many 0.1s?	Energy (J)
0.1	1	1 × 40 000 = 40 000
0.2	2	2 × 40 000 =
0.5		
1.5		

(b) Split the number. The first one is done for you.

Climb 2.5 floors: 2 floors + half a floor = 4 000 + 1 000 = **5 000** J

Climb 3.5 floors: 3 floors + half a floor = _____ + _____ = _____ J

Run 12.5 m: 12 m + half a metre = _____ + _____ = _____ J

(c) Mr Bowen's cup of tea needs 0.2 litres of water. How many **floors of stairs** could he climb with the energy that boils it? Use your answer to (a).

_____ ÷ 2 000 = _____ floors

X8 Mr Bowen's coffee goes cold

Mr Bowen makes a hot coffee. Then 30 students ask him questions. The room is **28.5 °C**.

Time (minutes)	0	10	20	30
Coffee (°C)	82.5	64.0	51.5	43.0

(a) How much does it cool in each 10 minutes? Subtract. The first one is done for you.

0 to 10 minutes

	8	2 .	5	
−	6	4 .	0	
	1	8 .	5	

10 to 20 minutes

	6	4 .	0	
−	5	1 .	5	
		.		

20 to 30 minutes

		.		
−		.		
		.		

(b) Each 10 minutes, does it cool by **more** or by **less**? Circle one. more less

(c) Which way does the **thermal energy** move? Tick one.

☐ from the coffee to the room ☐ from the room to the coffee

(d) Mr Bowen forgets his coffee for 3 hours. Will it get **colder** than the room? Circle one: yes no. Then explain in a full sentence. Start: **"Thermal energy only moves from …"**

Part 3 · Some angles

From Maths 5.1

Angles on a **straight line** add up to **180°**. Angles **around a point** add up to **360°**.
Two steps: **1.** Add the angles you know. **2.** Subtract the total from 180° or 360°.

X9 The Earth turns 15° every hour

The Earth spins once a day: 360° in 24 hours. 360 ÷ 24 = 15, so it turns **15° every hour**. Fill in the table.

Hours	Calculation	Degrees	At Mr Bowen's harbour
1	1 × 15	15°	
2	2 × 15		
6			high tide to low tide: a _____ angle
12			high tide to high tide: a _____ line
24			one whole day: a _____ turn
0.5	half of 15		
2.5	30 + _____		

X10 Equal angles around a point

Each one is cut into **equal** angles. Divide 360° by how many there are. Write one digit in each box.

(a) A wind turbine: 3 blades



$$3\overline{)360}$$

= _____°

(b) A ceiling fan: 5 blades



$$5\overline{)360}$$

= _____°

(c) A pizza: 8 slices



$$8\overline{)360}$$

= _____°

(d) A pizza: 16 slices



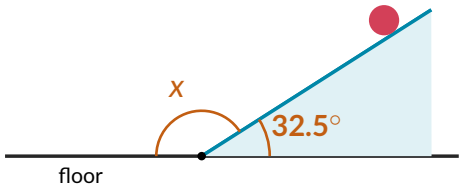
$$2\overline{)45.\square}$$

= _____°

(d) needs no new division: a 16-slice slice is **half** of an 8-slice slice. So it is (c)'s answer ÷ 2.

X11 The ramp

Mr Bowen rolls a ball down a ramp. The ramp makes 32.5° with the floor.



(a) Find x. Write 180 as 180.0.

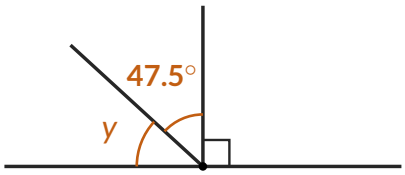
$$\begin{array}{r} 180.\square \\ -\ \ 32.5 \\ \hline \square\square\square.\square \end{array}$$

x = _____°

(b) The ball rolls down. Name the energy change.

_____ ⟶ _____

(c) Three angles on a straight line. Find y in two steps.



Step 1 add

$$\begin{array}{r} \ \ 90.\square \\ +\ \ 47.5 \\ \hline \square\square\square.\square \end{array}$$

Step 2 subtract

$$\begin{array}{r} 180.\square \\ -\ \square\square\square.\square \\ \hline \square\square\square.\square \end{array}$$

y = _____°

X12 The whole sheet in one question

Mr Bowen spends a day at the beach. High tide is **4.6 m** at **7 a.m.** Low tide is **0.75 m** at **1 p.m.**

(a) What is the **tidal range**?

Tidal range: _____ m

(b) From high tide to low tide is _____ hours.

The Earth turns _____ × 15 = _____°.

That is a _____ angle.

(c) Mr Bowen drops a ball from **2 m** onto the sand. It bounces only **0.35 m**. How far below 2 m does it stop?

_____ m

(d) In his hand, what energy store does the ball have?

_____

$$\begin{array}{r} \square\,4.6\square \\ -\ \square\,0.75 \\ \hline \square\square.\square\square \end{array}$$

$$\begin{array}{r} \square\,2.\square\square \\ -\ \square\,0.35 \\ \hline \square\square.\square\square \end{array}$$

(e) The ball did not come back to 2 m. Where did the rest of its energy go? Answer in a full sentence. Start: **"When the ball hit the sand, …"**

Answer Key — Teacher Copy

X1 Read the tide board

(a) 4 a.m. 5.4 · 6 a.m. 4.4 · 8 a.m. 2.2 · 10 a.m. 1.2 m (b) 4 a.m. (c) 10 a.m.

Each small line is 0.2, not 0.1. A student who counts ticks as tenths writes 5.2, 4.2, 2.1, 1.1: every reading half as far above the metre line. The remedy is the sheet's own line: count in 0.2s, 5, 5.2, 5.4.

X2 The tidal range

(b) 3.75 − 0.90 = 2.85 m (c) 6.10 − 2.35 = 3.75 m (d) 16.3 − 4.2 = 12.1 m

(b) and (c) are the orange-box zeros. Taking the smaller digit from the bigger in each column gives 3.25 for (b) and 4.25 for (c): the student has not borrowed.
(d) has no digits printed: the student places 16.3 and 4.2 in the grid. Both have one decimal place, so a wrongly placed grid can still give 12.1; check the grid, not only the answer.

X3 Half above, half below

4.5, 3.0: half 1.5 deepest 6.0 shallowest 3.0

3.2, 1.4: half 0.7 deepest 3.9 shallowest 2.5

6.0, 2.5: half 1.25 deepest 7.25 shallowest 4.75

(b) *He added the whole range to the average. He should add half of it:* 4.5 + 1.5 = 6.0 *m.*

The book's Question 6 (3.4), the trap the deck called out: the range is the *whole* distance from low to high.
Row 4 runs past the point: 2.5 ÷ 2 = 1.25. Expect 1.2 or 1.3 from a student who stops at one decimal place, and then 7.2 and 4.8.

X4 The land has tides too

(a) 0.3 m (b) 0.6 m (c) 42, 4.2 m
(d) *We do not feel it because everything goes up and down together: the ground, the buildings and us.*

(a) Expect 3 or 0.03: the point moved one place too few or one too many. 30 ÷ 100: two places left.
(c) 4.2 m in a week, and nobody noticed: that is the joke, and (d) is why.

X5 Six balls, one floor

(a) clay 1.95 · bouncy ball 0.25 · egg 2 (2.00) · basketball 0.65 · ping-pong 0.50
(b) 1.75, 1.5, 1.35, 1.1, 0.05, 0
(c) most: the egg (accept the clay) · least: the bouncy ball
(d) *The lower the bounce, the more energy changed to sound and heat at the floor.*

(a) is "write 2 as 2.00". An answer that starts with 1 (1.25 for the bouncy ball, 1.65 for the basketball) did not borrow from the 2. Counting up is the check: 1.75 + 0.25 = 2.
(b) 1.5 > 1.35 is the Unit 4 trap (the shorter number is bigger: 1.50). A list with 1.35 before 1.5 has compared the digit count.
(c) The egg and the clay both stop dead; either is right. The point is the direction: a *low* bounce means *more* energy left the ball.

X6 Drop it three times

(a) 1.12 + 1.08 + 1.13 = 3.33; 3.33 ÷ 3 = **1.11** m (b) 1.34 + 1.38 + 1.36 = 4.08; 4.08 ÷ 3 = **1.36** m

(b) has a 0 in the total. 4 ÷ 3 = 1 r 1, then 10 ÷ 3 = 3 r 1, then 18 ÷ 3 = 6. A student who drops the 0 divides 4.8 and gets 1.6 — bigger than every bounce, which is the check: a mean sits *between* the values.

X7 Joules

(a) 0.2: 80 000 · 0.5: 5 × 40 000 = 200 000 · 1.5: 15 × 40 000 = 600 000 J
(b) 3.5 floors: 6 000 + 1 000 = 7 000 J · 12.5 m: 2 400 + 100 = 2 500 J
(c) 80 000 ÷ 2 000 = 40 floors

(a) "How many 0.1s in 1.5?" is 15. Expect 5 (the digit after the point) or 1.5 (copied).
(b) Half a metre of running is 100 J; expect 200 from a student who reads "half a metre" as "a metre".
(c) is the deck's starter in small: the kettle in floors of stairs (there, 200 floors for a litre).

X8 Mr Bowen's coffee goes cold

(a) 64.0 − 51.5 = 12.5 · 51.5 − 43.0 = 8.5 (b) less (c) from the coffee to the room
(d) **no**. *Thermal energy only moves from hotter things to colder things, so the coffee stops cooling when it is as warm as the room (28.5 °C).*

(a) 64.0 − 51.5: expect 13.5, the smaller digit taken from the bigger in the tenths (5 − 0) instead of borrowing.
(d) is the reasoning question: the write note says which way heat moves, so once the coffee is at room temperature there is nowhere colder for it to go. "Yes, it keeps getting colder" has read the drops (18.5, 12.5, 8.5) as if they never stop.

X9 The Earth turns 15° every hour

2 h 30° · 6 h 90°, a **right** angle · 12 h 180°, a **straight** line · 24 h 360°, a **full** turn · 0.5 h 7.5° · 2.5 h 30 + 7.5 = 37.5°

Homework 11's F1 is this table's reason: the two bulges are 180° apart, so high tide comes every 12 hours.
0.5 × 15: expect 7 or 8 from a student who will not write a decimal angle.

X10 Equal angles around a point

(a) 120° · (b) 72° · (c) 45° · (d) 45.0 ÷ 2 = 22.5°

(b) and (c): 5 and 8 will not go into 3, so the first answer box is 0 (072, 045). A student who leaves the box empty and slides the digits left writes 720 or 450.
(d) runs past the point into the orange box: 45.0.

X11 The ramp

(a) 180.0 − 32.5 = 147.5° (b) gravitational potential → kinetic (c) 90 + 47.5 = 137.5; 180 − 137.5 = 42.5°

(a) Expect 152.5: the smaller digit taken from the bigger in the tenths and the ones, with no borrowing. The orange 0 is there to make the borrow visible.
(c) The small square is 90° and is not written as a number: expect 132.5° (180 − 47.5) from a student who missed it.

X12 The whole sheet in one question

(a) 4.60 − 0.75 = 3.85 m (b) 6 hours; $6 \times 15 = 90^\circ$; a right angle (c) 2.00 − 0.35 = 1.65 m
(d) gravitational potential
(e) *When the ball hit the sand, some of its energy changed to sound and heat, so it could not lift the ball back up to 2 m.*

(a) 4.6 needs its orange 0: expect 4.15 without it.
(e) "The sand is soft" is true and is not the answer; ask where the energy *went*.